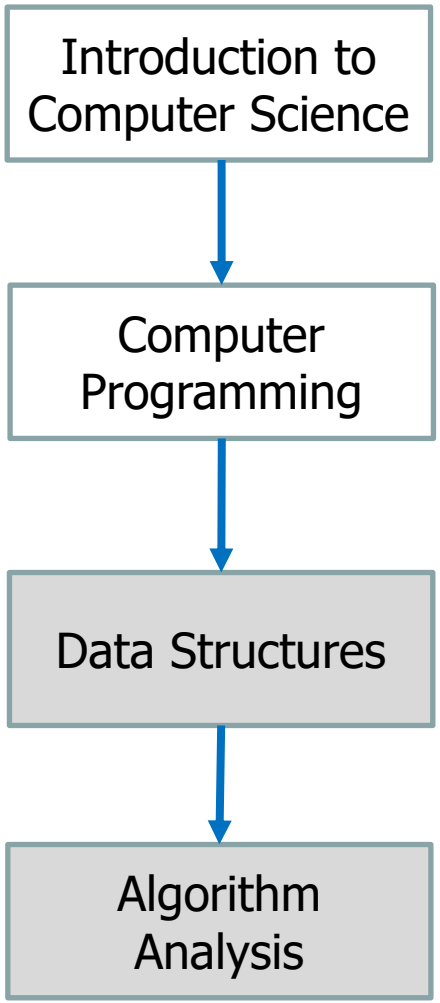
Data Structures
Fall 2023

National University of Computer and Emerging Sciences

General Overview



What is Hardware, Software, Programming, Operating System etc.

How to write software with the help of procedural and object oriented programming?

How to organize information/data, usually in memory, so that it can be used efficiently?

How to analyze the performance of algorithms to evaluate their suitability for solving complex problems?

Some Rules

- Raise your hand before asking any question
  - Wait for the permission
- Never ever miss a class
  - No retakes (except for Final Exam*)
- Never use mobile phone in the class

- Above all, whatever you do, please do not disturb others

* Conditional: as per university policy

Dishonesty and Plagiarism

- Any kind of cheating will be considered serious offense
- All parties involve in cheating will get minus (-) 50% marks
  - Exams, quizzes, assignments, projects
- Habitual cases will be awarded "F"

Course & Reference Books

- Data Structures Using C++ (By D.S. Malik)
- Data Structures Using C and C++ (By Y. Langsam, M. J. Augenstein, A. M. Tenenbaum)
- Data Structures and Algorithms (By A. V. Aho, J. E. Hopcroft, J. D. Ullman)
- Schaum's Outline Series, Theory and problems of Data Structures (By Seymour Lipschutz)

Attendance Policy

- Student arriving after the attendance will be marked "late"
  - No excuse to students arriving immediately after the attendance
- Two late arrivals will be treated as absent
- Students are not allowed to switch sections

What is a Data Structure?

- In a general sense, any representation that is used for storing information is a data structure
  - Example: integers, structures, classes, arrays, etc.

- More typically, a data structure provides a way for organizing a collection of data items
  - Storing, accessing and modifying data items

- Most data structures have associated algorithms to perform operations on data
  - Search, insert, remove etc.

Costs and Benefits

- Each data structure requires
  - Space for storing data items
  - Time to perform each basic operation
  - Programming efforts

- Cost of a data structure
  - Time and space resources in consumes

- Choice of data structure depends on many factors
  - Type of data
  - Frequency with which various data operations are applied

- Hard to define data structure that performs better in all situations
  - Time and space tradeoff

Goals of This Course

- Learn the commonly used data structures
  - Form a programmer's basic data structure "toolkit"
- Case studies of data structures
- Examine the costs and benefits of every data structure

Motivational Example

- A cellular service company provides contract to its 10 million users
- Due to new security enforcements, the company wants to prevent issuing of multiple contracts to users

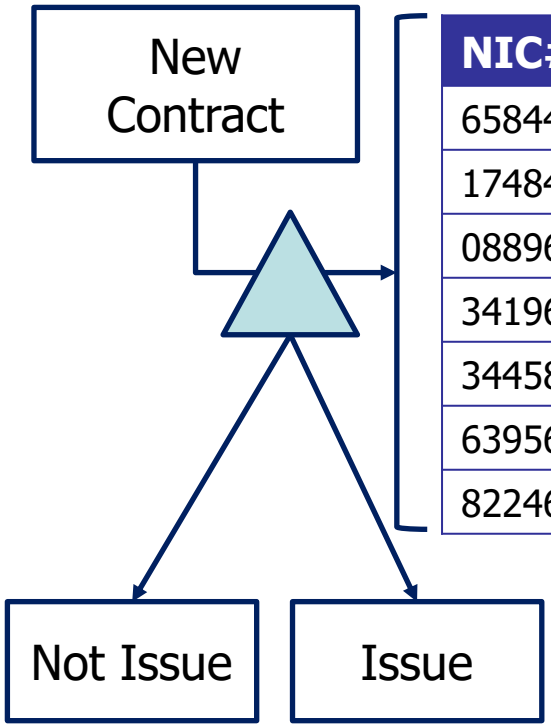
- Method of Detecting Multiple Contracts
  - Before issuing a new contract to user
  - First search the id of user in existing contracts database
  - In case of failure → issue a new contract
  - In case of success → do not issue a new contract to user

Example: Linear Array Data Structure (1)

NIC#	Name	Address
6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
0889679-1	Arslan Akhtar	H No 152 Bostang Colony
3419668-1	Zain Ahmed	Sharfabad Street Gulshan Karachi
3445864-3	Sumair Farooq	Post Office Tayyar, Multan
6395653-4	Ali Affan	H.No. 425, Sector F-11/4, Islamabad
8224641-1	Syed Faraz	Sharfabad Street Gulshan, Faisalabad

- Linear Array (with 10 million entries)
  - 3 arrays (NIC, Name, Address)
  - Structured array
  - Class's object array

Example: Linear Array Data Structure (2)



NIC#	Name	Address
6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
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- Any disadvantage of linear array data structure?
- How to improve?

Example: Improved Data Structure

- Create a dictionary data structure
  - Group similar records together
  - Similarity in terms of first digit of NIC number
  - Add a dictionary entry for each distinct digit (0 – 9)

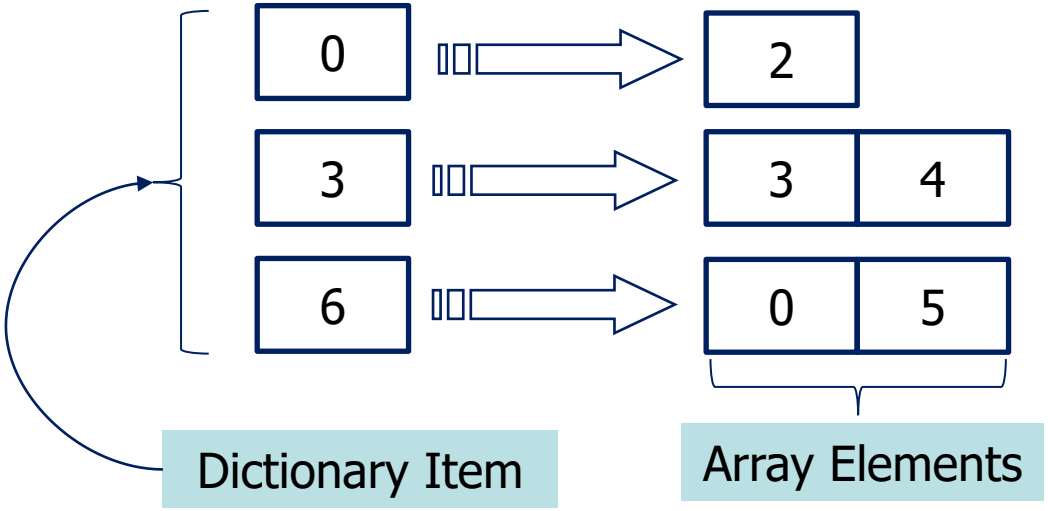
- Example: 3419668-1, 3445864-3, 1748425-5.
  - 3 and 1 are dictionary entries

- Existing contracts are searched in two steps
  1. Search the dictionary entry (i.e., group searching)
  2. Search contract within the group (i.e., with the same NIC digit)

- How much improvement w.r.t. linear array?

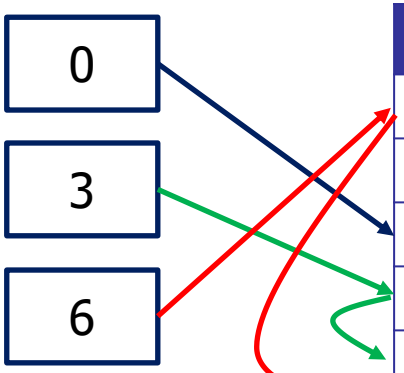
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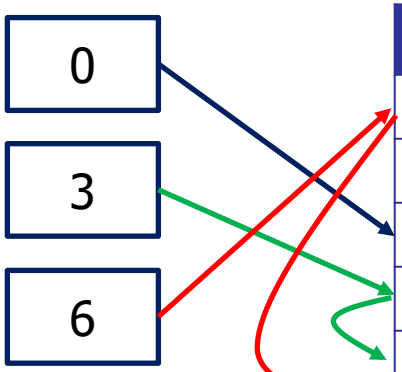


Example: Dictionary Data Structure (2)

- Another possibility
  - Maintain pointers with structures (or records)
  - Non NULL pointer indicates presence of next record



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Data Structure

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Where data structure is helpful?

- Data structure can make the difference between a program running in a few seconds or many days
  - Efficiency of data structure

What is Data Structure Efficiency?

- Two major measures of efficiency of a data structure or algorithm
  - Time and space
- Cost of data structure
  - Time and space resources it consumes
- Efficiency of a data structure depends on its operations
  - Makes a difference between program running in few seconds or days
- Difficult to define data structure that performs better in all situations

Any Question So Far?

